2.3 Waves

This topic covers the properties of different types of waves, including standing (stationary) waves. Refraction, polarisation and diffraction is also included.

This topic should be studied by exploring the applications of waves, for example, applications in medical physics or applications in music. This topic includes many opportunities to develop experimental skills and techniques.

Students will be assessed on their ability to:		Suggested experiments
28	understand and use the terms *amplitude*, *frequency*, *period*, *speed* and *wavelength*	Wave machine or computer simulation of wave properties
29	identify the different regions of the electromagnetic spectrum and describe some of their applications	
30	use the wave equation $v = f\lambda$	
31	recall that a sound wave is a longitudinal wave which can be described in terms of the displacement of molecules	Demonstration using a loudspeaker Demonstration using waves on a long spring
32	use graphs to represent transverse and longitudinal waves, including standing (stationary) waves	
33	explain and use the concepts of wavefront, coherence, path difference, superposition and phase	Demonstration with ripple tank
34	recognise and use the relationship between phase difference and path difference	
35	explain what is meant by a *standing (stationary) wave*, investigate how such a wave is formed, and identify nodes and antinodes	Melde's experiment, sonometer
36	recognise and use the expression for refractive index ${}_1\mu_2 = \sin i / \sin r = v_1/v_2$, determine refractive index for a material in the laboratory, and predict whether total internal reflection will occur at an interface using critical angle	
37	investigate and explain how to measure refractive index	Measure the refractive index of solids and liquids

Students will be assessed on their ability to:		Suggested experiments
38	discuss situations that require the accurate determination of refractive index	
39	investigate and explain what is meant by *plane polarised light*	Models of structures to investigate stress concentrations
40	investigate and explain how to measure the rotation of the plane of polarisation	
41	investigate and recall that waves can be diffracted and that substantial diffraction occurs when the size of the gap or obstacle is similar to the wavelength of the wave	Demonstration using a ripple tank
42	explain how diffraction experiments provide evidence for the wave nature of electrons	
43	discuss how scientific ideas may change over time, for example, our ideas on the particle/wave nature of electrons	
44	recall that, in general, waves are transmitted and reflected at an interface between media	Demonstration using a laser
45	explain how different media affect the transmission/reflection of waves travelling from one medium to another	
46	explore and explain how a pulse-echo technique can provide details of the position and/or speed of an object and describe applications that use this technique	
47	explain qualitatively how the movement of a source of sound or light relative to an observer/detector gives rise to a shift in frequency (Doppler effect) and explore applications that use this effect	Demonstration using a ripple tank or computer simulation
48	explain how the amount of detail in a scan may be limited by the wavelength of the radiation or by the duration of pulses	
49	discuss the social and ethical issues that need to be considered, e.g. when developing and trialing new medical techniques on patients or when funding a space mission	